PS-018

HAPS Link-Quality Prediction Using a Documented Channel Model

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Problem Description

HAPS link quality changes faster than operators can react, so service drops before action is taken.

- Rain, cloud and gas absorption weaken high-frequency HAPS links
- Platform drift, sway and jitter change distance and beam pointing over time
- Links are managed after SNR falls, not before, which causes outages
- PS-018 asks for ahead-of-time SNR and outage prediction from a documented channel model, tested on unseen conditions

Background

HAPS fly at about 20 km and are a key non-terrestrial layer for 6G, but their link budget depends on weather and motion.

- One stratospheric platform gives wide coverage, lower delay than satellites and faster deployment than towers [1]
- Non-terrestrial network (NTN) studies define path loss and elevation-based line of sight [2]
- ITU-R recommendations give rain, gas and cloud attenuation from weather inputs [3][4]

Link budget used to label data

$$SNR = EIRP + G/T - FSPL - A(rain) - A(gas) - A(cloud) - L(pointing) - k - B$$

Each loss term maps to a standard; platform drift and jitter change FSPL and pointing loss over time.

Literature & RFC Review

Only the most relevant standards and studies, each mapped to a module of our solution.

Source	What it provides	How we use it
[1] HAPS survey (Karapantazis & Pavlidou)	HAPS system and link overview	Scenario design
[2] 3GPP TR 38.811 (NTN)	NTN channel, path loss, LOS vs elevation	Propagation module
[3] ITU-R P.618	Earth-space link budget, scintillation	Fading and link budget
[4] ITU-R P.837 (+ P.676, P.840)	Rain, gas and cloud attenuation	Attenuation module (ITU-Rpy)
[5] XGBoost (+ GRU, SHAP)	Residual learning, sequence forecast, explanations	Hybrid predictor and explainer

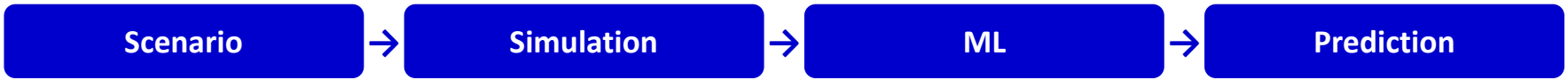
Proposed Solution

AI-powered link intelligence: a physics-based simulator feeding a hybrid ML predictor.

Predict SNR and outage risk several steps ahead

Explain which factors, such as rain or elevation, drive the drop

Act suggest beam, modulation or handover changes in time



Elevation, distance, band; rain, cloud, humidity; platform drift

Documented channel model; link budget and fading; labelled SNR series

Physics baseline + XGBoost residual; GRU forecasting

SNR for next steps; outage warning; suggested action

Optimization Proposed

Compared with a typical reactive approach, the proposed system changes five things.

Aspect	Before (typical)	After (proposed)
Timing	Reacts after the link drops	Predicts degradation in advance
Model	Black-box ML, hard to trust	Physics baseline + learned residual
Validation	Trained on one condition	Tested on unseen weather, bands and angles
Output	Reports a single number	SNR forecast, outage alert and action advice
Evidence	No clear reference point	Compared with persistence, analytical-only and ML-only baselines; SHAP for drivers

Solution Architecture & Design

Visual: classes of the link-intelligence engine and how data moves between them.

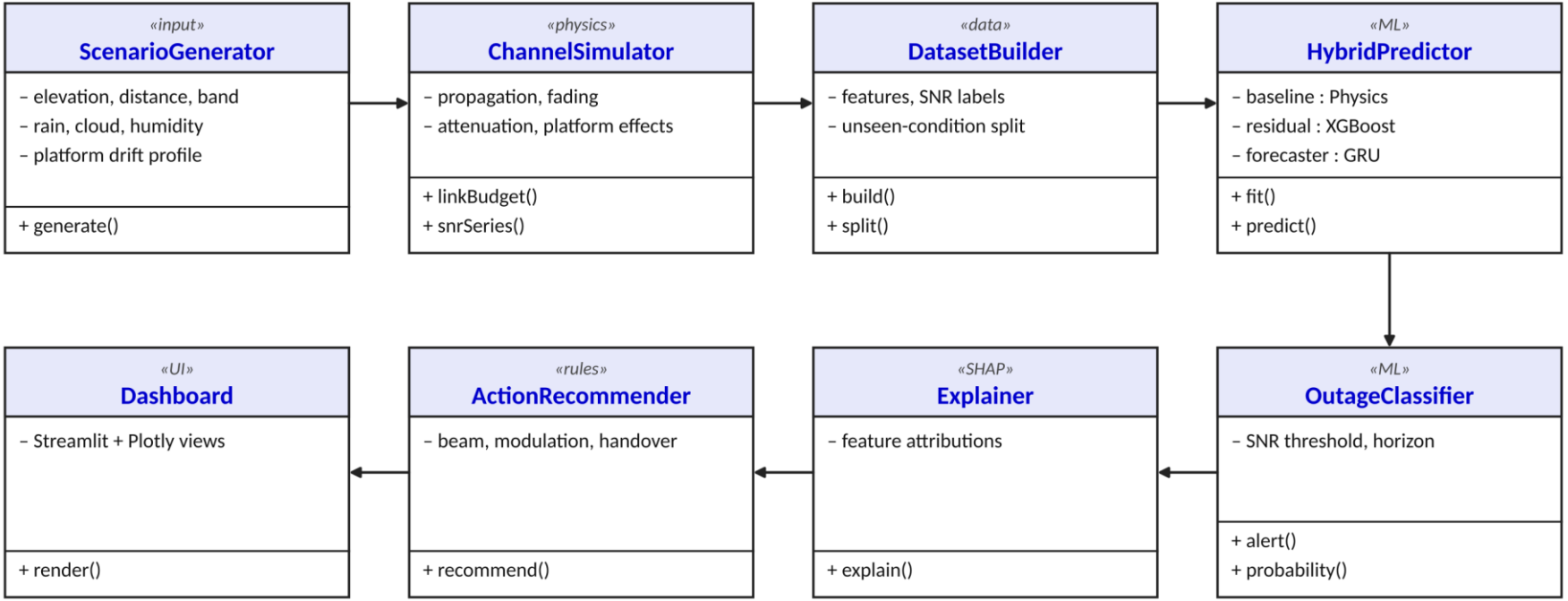


Figure: System Architecture UML Diagram

Data Flow Diagram

Overview of the architecture: external inputs, five processes and three data stores.

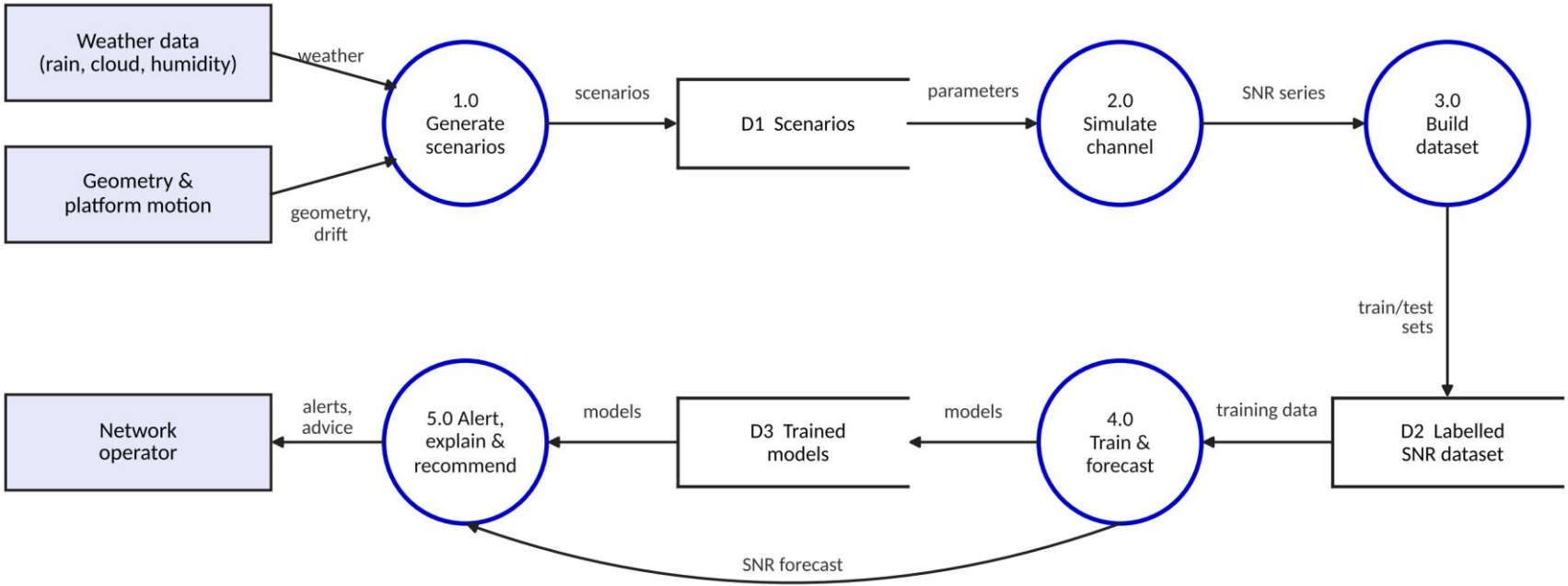


Figure: Data Flow Diagram (DFD)

Deployment Timeline

Four phases from idea to demo; each ends with a working, testable milestone.



Study & simulate

Review standards and build the channel simulator

Build dataset

Generate scenarios and set the three baselines

Train & forecast

Hybrid model, outage alerts and SHAP explainability

Demo & test

Dashboard, robustness tests and final report

Methodology: baseline-first and iterative; validate the physics model before adding ML; judge with RMSE, MAE, F1, recall and lead time; code in Jupyter and GitHub.

Bibliography I

[1] G. Karapantazis and F.-N. Pavlidou, “Broadband communications via high-altitude platforms: a survey,” IEEE Commun. Surveys & Tutorials, vol. 7, no. 1, pp. 2–31, 2005.
[2] 3GPP, “Study on New Radio (NR) to support non-terrestrial networks,” TR 38.811, v15.4.0, 2020.
[3] ITU-R, “Propagation data and prediction methods required for the design of Earth-space telecommunication systems,” Rec. P.618-14, 2023.
[4] ITU-R, “Characteristics of precipitation for propagation modelling,” Rec. P.837-7, 2017.
[5] T. Chen and C. Guestrin, “XGBoost: A scalable tree boosting system,” in Proc. ACM SIGKDD, 2016, pp. 785–794.

Thank You for Your Attention!

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“Instead of reacting after a HAPS link deteriorates, our system predicts link degradation in advance and provides actionable link intelligence for reliable 6G connectivity.”